BIHAR ENGINEERING UNIVERSITY, PATNA

DEPARTMENT OF ELECTRONICS & COMMUNICATION ENGINEERING

B.Tech 4th Semester • Examination Resource

ANALOG COMMUNICATION

Exam Date: Thursday, 29 Oct 2026 (2:00 PM - 5:00 PM)

Included Question Papers

Year	Paper Title	Course Code	Examination Session
2024	Analog Communication	104402	B.Tech 4th Semester Examination, 2024
2023	Analog Communication	104402	B.Tech 4th Semester Examination, 2023

Apna BEU Question Bank Archive • Compiled for B.Tech ECE 4th Semester

Contains official previous year questions with multiple-choice questions, descriptive problems, and complete circuits.

BIHAR ENGINEERING UNIVERSITY, PATNA

B.Tech 4th Semester Examination, 2024

ANALOG COMMUNICATION

Time: 03 Hours | **Code: 104402** | **Full Marks: 70**

Instructions:

1. The marks are indicated in the right-hand margin.
2. There are NINE questions in this paper.
3. Attempt FIVE questions in all.
4. Question No. 1 is compulsory.

Q.1 Choose the correct option / answer the following (Any seven question only):- **[14]**

1. If the carrier of a 100% modulated A.M. wave is suppressed, the power saving will be in percent-
   - i. 50
   - ii. 67
   - iii. 80
   - iv. 100

2. Modulation is the process of-
   - i. combining audio and radio frequency waves
   - ii. reducing distortion in RF amplifiers
   - iii. generating constant-frequency radio waves
   - iv. improving thermal stability of a system

3. The FM broadcast band is from-
   - i. 88 to 108 MHz
   - ii. 88 to 108 kHz
   - iii. 1600 kHz to 59.75 MHz
   - iv. 88 to 108 GHz

4. The standard intermediate frequency for FM is-
   - i. 10.7 MHz
   - ii. 10.7 kHz
   - iii. 4.5 MHz
   - iv. 4.5 kHz

5. With 100% modulation. ratio of sideband power to total power transmitted in an amplitude-modulated wave is-
   - i. 2/3
   - ii. 4/3
   - iii. 1/2
   - iv. 1/3

6. In an FM signal. as the modulation index increases. the power-
   - i. increases
   - ii. remains constant
   - iii. decreases
   - iv. None of above

7. The simplex system is a
   - i. bidirectional communication system
   - ii. multidirectional communication
   - iii. unidirectional communication
   - iv. None of the above

8. In _____ modulation the phase of the carrier signal is modulated by the message signal.
   - i. pulse
   - ii. amplitude
   - iii. phase
   - iv. frequency

9. In baseband signal the audio frequency signal range is around
   - i. 10-20 kHz
   - ii. 5-10 kHz
   - iii. 0-20 kHz
   - iv. 0-20 MHz

10. In super heterodyne receiver. at high frequency the
    - i. signal processing should be done
    - ii. signal transmission should be done
    - iii. bandwidth is low
    - iv. Both (i) and (ii)

Q.2 Solve both questions : **[14]**

1. Draw the block diagram of communication system. Explain the concept of frequency mixing.
2. Explain the function of each block of communication system.

Q.3 Solve both questions : **[14]**

1. Explain generation of DSB-SC signal with the help of balanced modulator using diodes.
2. Draw the frequency spectrum of DSB-SC modulation with necessary mathematical expressions. Write short note on Pre-Emphasis and De-Emphasis circuits.

Q.4 Solve both questions : **[14]**

1. Write short note on Pre-Emphasis and De-Emphasis circuits.
2. Draw block diagram of Super-heterodyne AM receiver and explain function of each block.

Q.5 Solve both questions : **[14]**

1. (i) The maximum peak-to-peak voltage of an AM wave is 12 mV and the minimum voltage is 4mV. Calculate the modulation factor. (ii) An audio signal of 1 KHz is used to modulate a carrier 500 KHZ. Determine side band frequencies and band width required.
2. Describe thermal noise and shot noise.

Q.6 Solve both questions : **[14]**

1. With block diagram explain the generation of PWM signals.
2. What are the differences between PAM, PWM, and PPM

Q.7 Solve both questions : **[14]**

1. The output signal from an AM modulator is: $s(t) = 5\cos(1800\pi t) + 20\cos(2000\pi t) + 5\cos(22000\pi t)$. (a) Determine the modulation index. (b) Determine the ratio of the power in the sidebands to the power in the carrier.
2. Explain (i) Signal to Noise Ratio (ii) Figure of merit.
   Discuss about noise effect in PM and obtain expression for figure of merit.

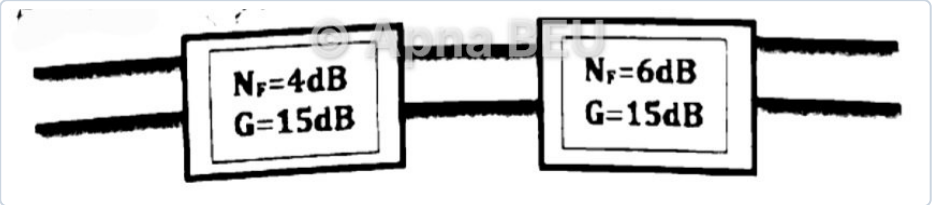
Q.8 Solve both questions : **[14]**

1. Define amplitude modulation and describe it in detail with suitable diagrams.
2. Write the difference forms of Amplitude Modulation, what is the efficiency of the AM?

Q.9 Solve both questions : **[14]**

1. Write short notes our Envelope Detector.

2. Find the overall noise factor for the cascaded noise networks as shown in given figure. [Stage 1: $N_F = 4dB$, $G = 15dB$; Stage 2: $N_F = 6dB$, $G = 15dB$]



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Q.1 Choose the correct answer of the following (Any seven question only): **[14]**

1. In NBFM, the modulation index is close to
   - i. 1
   - ii. 10
   - iii. greater than 1
   - iv. infinite

2. The maximum power efficiency of an AM modulator is
   - i. 50%
   - ii. 25%
   - iii. 75%
   - iv. 100%

3. Spectral density of white noise is
   - i. Varies with frequency
   - ii. Constant
   - iii. Varies with bandwidth
   - iv. Varies with amplitude

4. If the step size of a system is 0.0721, find its quantization power?
   - i. 3.25×10^{-3}
   - ii. 4.33×10^{-3}
   - iii. 4.33×10^{-4}
   - iv. 3.25×10^{-2}

5. Which of the following frequency is not transmitted in AM transmission?
   - i. Upper side band frequency
   - ii. Carrier frequency
   - iii. Lower side band frequency
   - iv. Audio frequency

6. What is the use of Companding?
   - i. in PCM transmitters to allow amplitude limiting in the receivers
   - ii. in PCM receiver to overcome impulse noise
   - iii. to overcome quantizing noise in PCM
   - iv. to protect small signals in PCM from quantizing distortion

7. Envelope Detector is a/an
   - i. Coherent detector
   - ii. Asynchronous Detector
   - iii. Synchronous Detector
   - iv. Product Demodulator

8. A 1000KHz carrier is modulated by 100Hz and 200Hz waves, then which of the following frequencies cannot be present at the output?
   - i. 1000.1 KHz
   - ii. 999.9 KHz
   - iii. 1000.2 KHz
   - iv. 999.7 KHz

9. For 100% modulation, power in each sideband is ____ of that of carrier.
   - i. 50%
   - ii. 60%
   - iii. 25%
   - iv. 33.33%

10. The frequency of local oscillator
    - i. can be either below or above the RF frequency
    - ii. is below the RF frequency
    - iii. is above the RF frequency
    - iv. is fixed typically at 450KHz

Q.2 Solve this question : **[14]**

1. A radio receiver with equivalent noise bandwidth of 10kHz has a noise figure of 20dB.If input SNR to receiver is 40dB, determine the output SNR. What is equivalent noise temperature if ambient temperature is 27°C.

Q.3 Solve this question : **[14]**

1. Discuss the Armstrong method for the generation of wideband FM.

Q.4 Solve both questions : **[14]**

1. Explain the working of slope detector method used for demodulation.
2. Write notes on PAM, PPM and PWM.

Q.5 Solve both questions : **[14]**

1. Explain need of modulation and its advantage.
2. Derive the expression for power content in an AM wave.

Q.6 Solve this question : **[14]**

1. When a super-heterodyne receiver is tuned to 600kHz,its local oscillator provides the mixer with an input at 1055 kHz. What is the image frequency? The antenna of the receiver is connected to the mixer via a tuned circuit with Q 50. What will be the rejection ratio for the calculated image frequency?

Q.7 Solve this question : **[14]**

1. What is demodulation? Explain square law demodulator.

Q.8 Solve this question : **[14]**

1. Derive the expression of AM modulated wave and draw its waveform.

Q.9 Write short notes on any two of the following: **[14]**

a. Thermal Noise and Shot Noise & Carson Rule.

b. DSB-SC & VSB

c. Difference between Analog & Digital communication

d. Synchronous detection of DSB-SC